

[← Blog](#)

Blog

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Kazakhstan AI/ML community is delighted with VeriMeet

At Verigram, we believe that the perspectives of AI and ML aren't just about algorithms and data. It's about fostering a vibrant community of professionals who share a passion for innovation and technology.

On September 8th, we took a significant step in that direction by hosting our very first AI/ML community gathering in Almaty, and the results were nothing short of extraordinary.

Resounding Success

Our meeting room brought together decision makers, computer vision experts, and machine learning engineers representing renowned organizations from various industries, including banking, telecommunications, retail, and consulting.

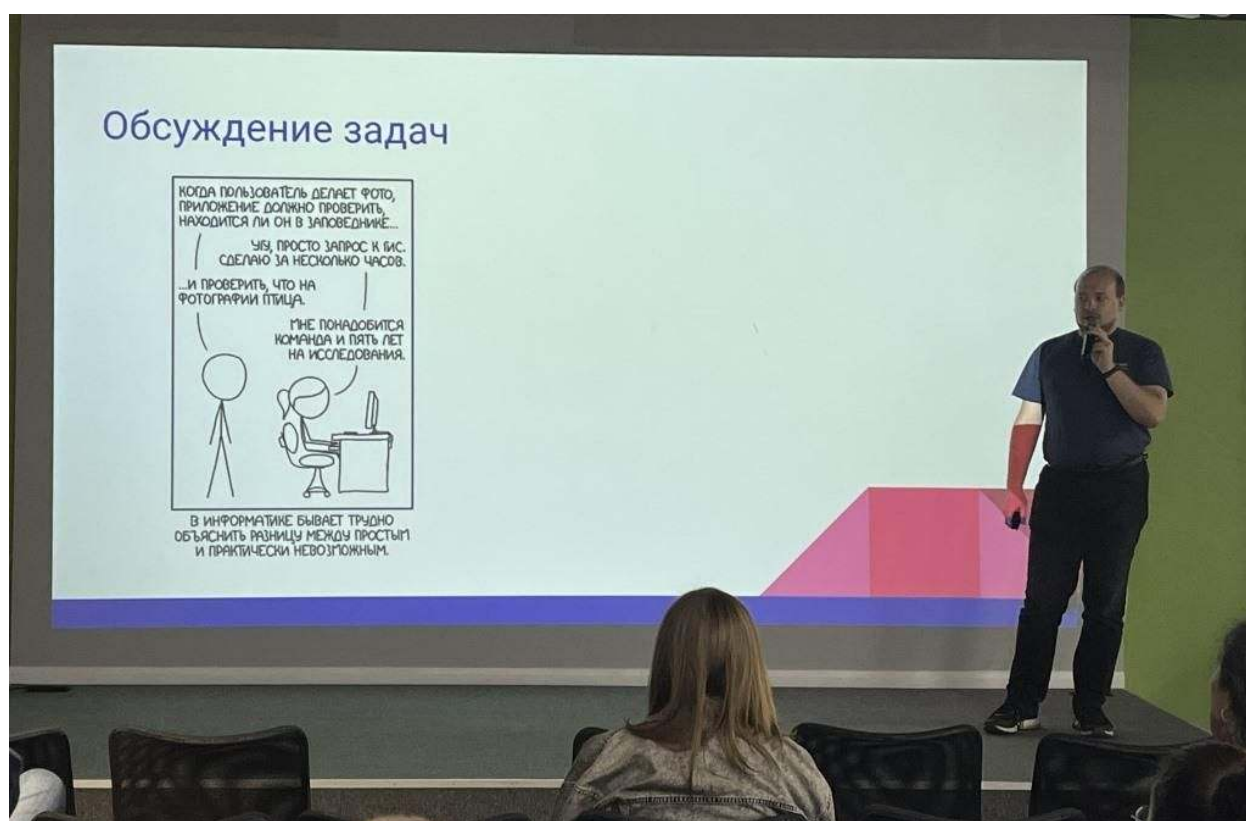


The central theme of the event, «Machine Learning Today — Difficult, Expensive, but Hellishly Interesting», piqued the interest of all present. As a leading provider of biometric verification solutions, we were excited to present our ideas on facial video analysis and discuss how to improve collaboration between business and research teams. We were met with great enthusiasm and engagement from the audience, the Q&A sessions seemed endless, and the chats during the coffee breaks turned into lively discussions.

Conference speakers highlighted cutting-edge engineering data and ways to improve communication between business and research methodologies. Reports on developments in computer vision and the use of robot assistant voices were of great interest. The basic presentations were based on real cases and were rich in technical details.

Researchers and business: better understanding and cooperation

Andrey Shadrikov, head of the R&D group at Verigram, talked about the importance of «speaking the same language» as the business. How to avoid contradictions and disappointments, how to synchronize development expectations, how to keep researchers from bypassing business



«What happens when you talk to business? The business asks questions, and R&D comes up with phrases that mean nothing to them. There will be improvements — this is just one of those phrases,» Andrey points out the problem. The presentation was particularly useful for developers and researchers. The speaker gave practical recommendations on how to communicate with a business team, how to build a scope of responsibility and how to make the management understand what the end result will be.

What is needed for this?

- To use clear metrics;
- To exchange customer experience;
- To agree on expectations;
- To save research results.

Such an approach makes communications much more transparent and does not expose either party.



project, but **Kadmir Kadyrov**, CV engineer of Verigram, thinks otherwise.

«There's a lot that can go wrong when building a system like this. One of the big issues we've encountered is the quality of the face images. Faces can be too bright, blurry, at an unusual angle, or completely obscured».



What should a Computer Vision engineer pay attention to? Three fundamental parameters:

- the camera;
- the environment;
- the faces.

A lot depends on the camera resolution, lens quality, weather and lighting conditions, location, pose, distance of the face to the camera. The better you want to make the system, the deeper you should dive into how to handle these external factors.

Data sets for Computer Vision: Expectancy does not equal reality

How different is data from the Internet from data in the real world? What should a computer vision engineer be prepared for before starting to work with raw data? **Dmitriy Gordin**, a Computer Vision engineer of Citix, outlined real challenges in the work of an engineer when working with urban gadgets that analyze the behavior of people and cars.



Seemingly real platforms like Kaggle, where people compete on CV/ML tasks and have access to datasets, actually have little in common with real-world data. People, cars, and external conditions are as unpredictable as possible. In addition, business requests are also added to this.

Oddly enough, synthetic data and AI tools are very useful for solving such problems. Data is the key!

- Combine all the datasets that you can download;
- Utilize pseudo-labeling with good pre-trained model;
- Apply heavy augmentations;
- Active learning to expand dataset with real-data.

Voice Robots: multilingualism and mixed language

Smart voice robots around the world are optimizing costs for businesses while creating new opportunities for help desks. Technology alone creates many new businesses and jobs, as all these systems must be improved and maintained. **Aigerim Kambetbayeva**, a Machine Learning specialist of Cybernet Kazakhstan, shared a report about their company's journey to develop a smart voice robot.



This technology is expected to have:

- high performance and scalability;
- high recognition accuracy;
- multilingualism (in the case of Kazakhstan, even the ability to work with combined speech);
- security and confidentiality.

The speaker assured: «The dataset is 90% of the success of any model. In the case of Kazakhstan, collecting 300 hours of Russian-Kazakh audio data was not an easy task». Through trial and error, the Cybernet Kazakhstan team achieved from zero to 3 million conversations per month in 2 years.

How to handle a billion new records per day when migrating databases

Large companies are willing to pay huge amounts of money to effectively manage data flows. Today there are many data migration practices and big data management frameworks. One of such well-known DBMS is ClickHouse. **Kirill Markin**, an experienced Data Scientist, shared a case study of how he and his team coped with data migration, with a load of a billion new records daily.



What conclusions did Kirill come to as a result?

- Before migrating old data, you should conduct an audit and double-check everything;
- If you have an analytical storage, then flat data in a raw single event table can be a good idea;
- Migration is dozens of tasks for 3 employees;
- Migration is expensive and time-consuming.

AI technology in Big tech corporations

Big tech corporations such as ByteDance (TikTok) and Meta are incorporating AI technology into their products to varying degrees. In the case of TikTok, the entire business is tied to a recommendation system, which directly depends on the results of training machine learning models. At the same time, Meta wants to start embedding AI into every single product. Banks are also actively implementing machine learning technologies in the areas of credit scoring and customer support. At the panel discussion, we were lucky enough to ask questions of experts who have worked with these top companies.



«ByteDance has been working hard on its recommendation system since its inception 12 years ago. When you talk to clients and tell them the dimensions of a model that is hundreds of terabytes, they are in great shock.» — says **Aibek Sarimbekov**, Director of Sales Engineering at ByteDance — «the model can recommend anything: text, video. You can make a new Spotify, a news aggregator, everything is possible».

What about machine learning at Blockchain.com, one of the most famous Bitcoin block explorers and cryptocurrency wallet services? **Zhanibek Kaimuldenov**, an Engineering Manager of Blockchain.com, shares: «Machine learning is actively used in customer support: clustering, ticket routing. Before this, some employees did not understand how to create tickets, to whom they were assigned, etc., but after the implementation of ML the situation changed for the better».

Sharing Expertise and Networking

The conference participants presented various aspects of AI/ML. They included technical details, real cases, conclusions from experiments and tests, and covered a wide range of application areas. Engineers from leading companies such as CEREBRA AI and Bereke Bank participated in the discussions.



is growing. Such meetings as VeriMeet arouse interest not only among engineers, but also among business representatives. Here you can hear new ideas, solve complex topical problems, demonstrate your achievements and simply communicate on exciting topics of technology development. We were happy to see everyone and look forward to more meetings. The Verigram team is open to new ideas and suggestions.

If you attended our event, [please answer a few questions](#). This will help us make the next event even better and more interesting.

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